

MAT1002 Lecture 3,

Outline

- Comparison tests (10.4)
- Absolute convergence (10.5)
- Ratio test
- Root test
- Alternating series (10.6)

Comparison Tests

Theorem ((Direct) Comparison Test)

Let $\sum a_n$ and $\sum b_n$ be series. Suppose that there exists an integer N such that $a_n \geq b_n \geq 0$ for all n satisfying $n \geq N$.

- (i) If $\sum a_n$ converges, then $\sum b_n$ converges.
- (ii) If $\sum b_n$ diverges, then $\sum a_n$ diverges.

Proof: (i) Suppose $\sum_{n=N}^{\infty} a_n = L$. For $n \geq N$, let

$$S_n := \sum_{k=N}^n b_k = b_N + \dots + b_n.$$

Then for all $n \geq N$,

$$0 \leq S_n = \sum_{k=N}^n b_k \leq \sum_{k=N}^n a_k \leq L.$$

Hence $\{S_n\}$ is bounded, implying that $\sum_{n=N}^{\infty} b_n$ converges by the Monotonic Sequence Theorem.

(ii) Equivalent to (i). □

E.g. Determine the convergence of $\sum_{n=8}^{\infty} \frac{1}{3^n + n^{1/3}}$.

Sol. $\forall n \geq 8, \quad 0 < \frac{1}{3^n + n^{1/3}} < \left(\frac{1}{3}\right)^n.$

Since $\sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n$ converges (geometric series with $|r| < 1$),

Series converges by (direct) comparison test.

e.g. The series $\sum_{n=1}^{\infty} \frac{\ln n}{n}$ diverges, since $0 < \frac{1}{n} < \frac{\ln n}{n}$

$\forall n \geq 3$, and $\sum \frac{1}{n}$ diverges.

proof: (i) $\exists N_0$ st $\forall n \geq N_0$

$$\left| \frac{a_n}{b_n} - L \right| = \frac{L}{2} (= \epsilon)$$

$$\Rightarrow -\frac{L}{2} < \frac{a_n}{b_n} - L \leq \frac{L}{2}$$

$$\frac{L}{2} b_n < a_n = \frac{3}{2} L b_n$$

For $n \geq \max(N, N_0)$

if $\sum a_n$ converges

$\sum \frac{1}{2} b_n$ converges

Theorem (Limit Comparison Test)

Let $\sum a_n$ and $\sum b_n$ be series. Suppose that there exists an integer N such that $a_n > 0$ and $b_n > 0$ for all n satisfying $n \geq N$.

(i) If $\lim_{n \rightarrow \infty} (a_n/b_n) = L > 0$, then $\sum a_n$ converges if and only if $\sum b_n$ converges. ($\in \mathbb{R}$)

$\rightarrow b_n > a_n$

(ii) If $\lim_{n \rightarrow \infty} (a_n/b_n) = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.

(iii) If $\lim_{n \rightarrow \infty} (a_n/b_n) = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

$\rightarrow a_n > b_n$

Proof: (i) To be shown in class. (ii) & (iii): exercise.



e.g. (a) $\sum_{n=1}^{\infty} \frac{\ln n}{n^{5/4}}$ convs or not?

Sol. Compare with $\frac{1}{n^{4.5/4}} = \frac{1}{n^{9/8}}$.

$$\frac{\ln n}{n^{5/4}} / \frac{1}{n^{9/8}} = (\ln n) n^{9/8} n^{-5/4} = \frac{\ln n}{n^{1/4} n^{1/8}}$$

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n^{1/4} n^{1/8}} \stackrel{L'Hô}{=} \lim_{n \rightarrow \infty} \frac{1/n}{\frac{1}{4} n^{-3/4} + \frac{1}{8} n^{-7/8}} = \lim_{n \rightarrow \infty} \frac{1}{4 n^{1/4} + 8 \frac{1}{n^{1/8}}} = 0$$

Since $\sum_{n=1}^{\infty} \frac{1}{n^{9/8}}$ converges (p-series with $p = 9/8$),

$\sum_n \frac{\ln n}{n^{5/4}}$ converges.

(b) $\sum_{n=3}^{\infty} \sin \frac{1}{n}$ cys or not?

$$\lim_{n \rightarrow \infty} \frac{\sin \frac{1}{n}}{\frac{1}{n}} = 1 \quad \text{share the same convergence}$$

$$\sum_{n=3}^{\infty} \frac{1}{n} \rightarrow \text{diverges so}$$

Now we shift our attention to series with both positive and negative terms, for which integral test and comparison tests may not work. $\sum_{n=3}^{\infty} \sin \frac{1}{n}$ diverges.

Absolute Convergence

Definition

A series $\sum a_n$ is said to converge absolutely if $\sum |a_n|$ converges.

e.g. $\sum_{n=1}^{\infty} \frac{\sin n}{n^2}$ Converges absolutely. Indeed,

$$0 < \left| \frac{\sin n}{n^2} \right| < \frac{1}{n^2}, \quad \forall n \geq 1 \quad (n \in \mathbb{Z}),$$

So $\sum_n \left| \frac{\sin n}{n^2} \right|$ converges by comparison test.

Theorem (Absolute Convergence Test)

If $\sum |a_n|$ converges, then $\sum a_n$ converges. In other words, if $\sum a_n$ converges absolutely, then it converges.

Proof: Since $a_n + |a_n| = \begin{cases} 0, & \text{if } a_n < 0 \\ 2|a_n|, & \text{if } a_n \geq 0 \end{cases}$

$$0 \leq a_n + |a_n| \leq 2|a_n|, \quad \forall n.$$

If $\sum |a_n|$ converges, then $\sum (a_n + |a_n|)$ converges, by comparison test.

This means $\sum a_n = \sum (a_n + |a_n|) - \sum |a_n|$ also converges. $\square$

Remark: The converse is false. As we will see, the alternating harmonic series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ converges, although $\sum_{n=1}^{\infty} |(-1)^{n+1} \frac{1}{n}| = \sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

The ratio test and root test can be used to test for absolute convergence.

Theorem (Ratio Test)

Let $\sum a_n$ be a series, and suppose that

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L \in \mathbb{R} \cup \{\infty\}.$$

Then:

- (i) if $L < 1$, the series converges absolutely.
- (ii) if $L > 1$ or $L = \infty$, the series diverges.
- (iii) if $L = 1$, the test is inconclusive.

e.g. (a) $\sum_{n=1}^{\infty} \frac{n!}{n^n}$

$$\text{Since } \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \frac{(n+1)}{(n+1)} \cdot \frac{n^n}{(n+1)^n} = \left(\frac{n}{n+1}\right)^n = \frac{1}{\left(1+\frac{1}{n}\right)^n} = \frac{1}{e} < 1,$$

$$\left| \frac{a_{n+1}}{a_n} \right|$$

$\sum_n \frac{n!}{n^n}$ converges absolutely.

$$(b) \sum_{n=1}^{\infty} \frac{2^n + 5}{3^n}$$

$$\text{Since } \frac{2^{n+1} + 5}{3^{n+1}} \cdot \frac{3^n}{2^n + 5} = \frac{2^{n+1} + 5}{3(2^n + 5)} = \frac{1 + \cancel{5 \cdot 2^{-n-1}}}{3(2^{-1} + \cancel{5 \cdot 2^{-n-1}})} \rightarrow \frac{2}{3} < 1, \text{ as } n \rightarrow \infty$$

$\sum_n \frac{2^n + 5}{3^n}$ converges absolutely.

Proof of Ratio Test

note that this is O.K.

(ii) If $L > 1$ or $L = \infty$, then $\exists N$ s.t. $\forall n \geq N$,

$$\left| \frac{a_{n+1}}{a_n} \right| > 1, \text{ i.e., } |a_{n+1}| > |a_n|, \text{ so}$$

$$0 \leq |a_n| < |a_{n+1}| < |a_{n+2}| < |a_{n+3}| < \dots$$

This means that $\lim_{n \rightarrow \infty} a_n \neq 0$, so $\sum a_n$ diverges.

(i) If $L < 1$, let $r \in (L, 1)$. Then $\exists N$ s.t. $\forall n \geq N$,

$$\left| \frac{a_{n+1}}{a_n} \right| < r. \text{ This means}$$

$$|a_{n+1}| < r |a_n|$$

$$|a_{n+2}| < r |a_{n+1}| < r(r |a_n|) = r^2 |a_n|$$

$$\vdots$$

$$|a_{n+k}| < r |a_{n+k-1}| < r(r^{k-1} |a_n|) = r^k |a_n|$$

$$\text{Since } 0 < |a_{n+k}| \leq r^k |a_n| \quad \forall k \geq 0$$

and $\sum_{k=0}^{\infty} |a_n| r^k$ converges ($0 \leq L < r < 1$), we have

$$\sum_{k=0}^{\infty} |a_{n+k}| = \sum_{n=n+k}^{\infty} |a_n| \text{ converges.}$$

(iii) For a p -series $\sum_{n=1}^{\infty} \frac{1}{n^p}$, $\left| \frac{a_{n+1}}{a_n} \right| = \frac{n^p}{(n+1)^p} \rightarrow 1^p = 1$ as

$n \rightarrow \infty$. However, it may converge (say $p=2$) or diverge (say $p=1$).

Theorem (Root Test)

Let $\sum a_n$ be a series, and suppose that

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L \in \mathbb{R} \cup \{\infty\}.$$

Then:

- (i) if $L < 1$, the series converges absolutely.
- (ii) if $L > 1$ or $L = \infty$, the series diverges.
- (iii) if $L = 1$, the test is inconclusive.

e.g. (a) $\sum_{n=1}^{\infty} a_n$, where $a_n = \begin{cases} n/2^n, & \text{if } n \text{ is odd.} \\ 1/2^n, & \text{if } n \text{ is even.} \end{cases}$

• Note that it is not convenient to use the ratio test.

• Since

$$\sqrt[n]{|a_n|} = \begin{cases} \frac{\sqrt[n]{n}}{2}, & \text{if } n \text{ is odd} \\ \frac{1}{2}, & \text{if } n \text{ is even} \end{cases}, \quad n \rightarrow \infty$$

$\frac{1}{2} \leq \sqrt[n]{|a_n|} \leq \frac{\sqrt[n]{n}}{2}$, $\forall n$. Since $\frac{\sqrt[n]{n}}{2} \rightarrow \frac{1}{2}$, by Sandwich theorem,

$\sqrt[n]{|a_n|} \rightarrow \frac{1}{2} < 1$. By root test, $\sum a_n$ convs absolutely.

$$(b) \sum_{n=1}^{\infty} \frac{2^n}{n^3}$$

$n\sqrt[n]{|a_n|} = \frac{2}{(\frac{1}{n})^3} \rightarrow 2$ as $n \rightarrow \infty$, so series diverges.

Proof: (ii) If $L > 1$ or $L = \infty$, then $\exists N$ s.t. $\forall n \geq N$,

$$n\sqrt[n]{|a_n|} > 1 \Rightarrow |a_n| > 1^n = 1 \Rightarrow \lim_{n \rightarrow \infty} a_n \neq 0.$$

So $\sum_n a_n$ diverges.

(i) If $L < 1$, let $r \in (L, 1)$. Then $\exists N$ s.t. $\forall n \geq N$,

$$n\sqrt[n]{|a_n|} < r \Rightarrow |a_n| < r^n. \text{ Since } 0 \leq |a_n| < r^n, \forall n \geq N,$$

and $\sum_{n=N}^{\infty} r^n$ converges, by comparison test, $\sum |a_n|$ converges.

(iii') Again, for any p -series $\sum \frac{1}{n^p}$, $(|a_n|)^{\frac{1}{n}} = \frac{1}{(n^{\frac{1}{n}})^p} \rightarrow 1$ as $n \rightarrow \infty$. But the p -series may converge ($p=2$) or diverge ($p=1$).



Alternating Series

Definition

An **alternating series** is a series of the form $\sum (-1)^{n+1} u_n$, where $u_n > 0$ for all n .

Remark

The starting index does not have to be 1 (so the first term can be negative). For example, both

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \quad \text{and} \quad -1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \dots$$

are alternating series.

10.6.15

THEOREM 15—The Alternating Series Test The series

$$\sum_{n=1}^{\infty} (-1)^{n+1} u_n = u_1 - u_2 + u_3 - u_4 + \cdots$$

converges if all three of the following conditions are satisfied:

1. The u_n 's are all positive.
2. The positive u_n 's are (eventually) nonincreasing: $u_n \geq u_{n+1}$ for all $n \geq N$, for some integer N .
3. $u_n \rightarrow 0$.

e.g. The alternating p -series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^p}$ converges for all $p > 0$, since:

- $\{\frac{1}{n^p}\}$ positive and $\searrow$
- $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$.

In particular, the alternating harmonic series $\sum (-1)^{n+1} \frac{1}{n}$ convs.

Fact: If $\{a_1, a_3, a_5, \dots\}$ and $\{a_2, a_4, a_6, \dots\}$ both converge to the limit L , then so does $\{a_1, a_2, a_3, a_4, \dots\}$, i.e., $\lim_{n \rightarrow \infty} a_n = L$.

Proof: Exercise. (10.1, ex 133.)

Proof (of the Alternating series test): (For notational simplicity, assume $N=1$.)

- Let S_n be the partial sum. Consider $b_k := S_{2k}$.
- For any $k \geq 1$,

$b_k = S_{2k} = (u_1 - u_2) + (u_3 - u_4) + \dots + (u_{2k-1} - u_{2k})$,
 so $\{b_k\}$ is nondecreasing, since $u_{2i-1} - u_{2i} \geq 0$, $\forall i$.

• $\{b_k\}$ is bounded, since

$$\begin{aligned} 0 &\leq (u_1 - u_2) + \dots + (u_{2k-1} - u_{2k}) \\ &= u_1 - (u_2 - u_3) - (u_4 - u_5) - \dots - (u_{2k-2} - u_{2k-1}) - u_{2k} \\ &\leq u_1. \end{aligned}$$

• By Monotonic Sequence Theorem, $\{b_k\} = \{S_{2k}\}$ converges,
 say $S_{2k} \rightarrow L$.

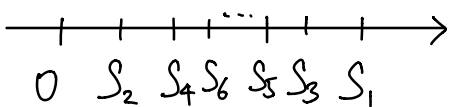
• Consider odd sequence $\{S_{2k-1}\}$. Then

$$S_{2k} = S_{2k-1} - u_{2k} \Rightarrow S_{2k-1} = S_{2k} + u_{2k}$$

$$\Rightarrow \lim_{k \rightarrow \infty} S_{2k-1} = \lim_{k \rightarrow \infty} S_{2k} + \lim_{k \rightarrow \infty} u_{2k} = L + 0 = L.$$

• By Fact before proof, $\lim_{n \rightarrow \infty} S_n = L$, so series converges.



Intuition:  $|S_n - S_{n-1}| = u_n \rightarrow 0$

Remark:

If we have an alternating series that starts with a negative term, may multiply the series by -1 , then use the test.

e.g.,

$$-1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \dots = -\left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots\right).$$

e.g. For $\sum (-1)^{n+1} \frac{n^2}{n^3+1}$:

- $f(x) := \frac{x^2}{x^3+1} \searrow$ on $[\sqrt[3]{2}, \infty)$

- Cvg by alternating series test.